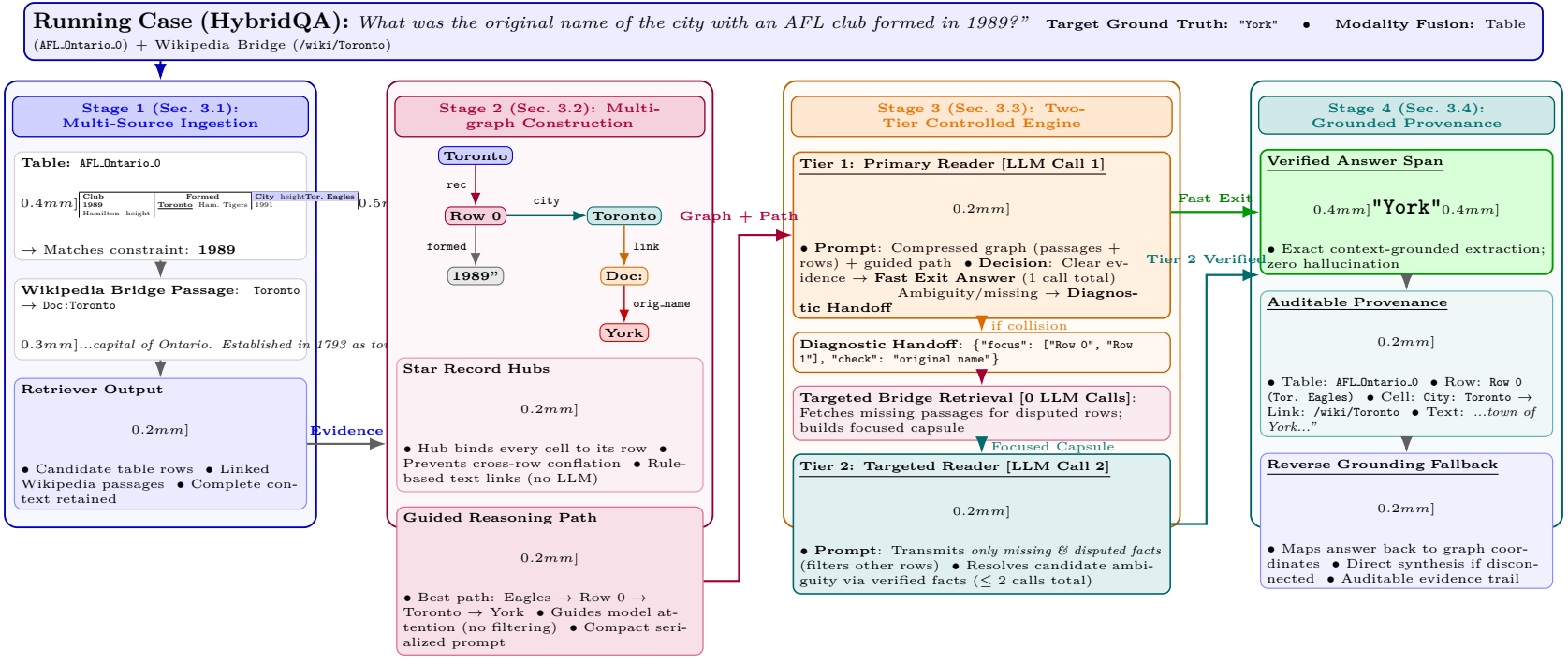

Running Case (HybridQA): What was the original name of the city with an AFL club formed in 1989?" Target Ground Truth: "York" • Modality Fusion: Table (AFL_Ontario_0) + Wikipedia Bridge (/wiki/Toronto)
Stage 1 (Sec. 3.1): Multi-Source Ingestion
Table: AFL_Ontario_0
Club
1989
Hamilton
Formed
Toronto
Ham. Tigers
City
1991
Tor. Eagles
→ Matches constraint: 1989
Wikipedia Bridge Passage: Toronto → Doc:Toronto
...capital of Ontario. Established in 1793 as to
Retriever Output
• Candidate table rows • Linked Wikipedia passages • Complete context retained
Evidence
Stage 2 (Sec. 3.2): Multi-graph Construction
Toronto
rec
Row 0
city
Toronto
formed
1989"
link
Doc:
orig_name
York
Graph + Path
Star Record Hubs
• Hub binds every cell to its row • Prevents cross-row conflation • Rule-based text links (no LLM)
Guided Reasoning Path
• Best path: Eagles → Row 0 → Toronto → York • Guides model attention (no filtering) • Compact serialized prompt
Stage 3 (Sec. 3.3): Two-Tier Controlled Engine
Tier 1: Primary Reader [LLM Call 1]
• Prompt: Compressed graph (passages + rows) + guided path • Decision: Clear evidence → Fast Exit Answer (1 call total) Ambiguity/missing → Diagnostic Handoff
Fast Exit
if collision
Diagnostic Handoff: {"focus": ["Row 0", "Row 1"], "check": "original name"}
Targeted Bridge Retrieval [0 LLM Calls]: Fetches missing passages for disputed rows; builds focused capsule
Focused Capsule
Tier 2: Targeted Reader [LLM Call 2]
• Prompt: Transmits only missing & disputed facts (filters other rows) • Resolves candidate ambiguity via verified facts (≤ 2 calls total)
Tier 2 Verified
Stage 4 (Sec. 3.4): Grounded Provenance
Verified Answer Span
"York"
• Exact context-grounded extraction; zero hallucination
Auditable Provenance
• Table: AFL_Ontario_0 • Row: Row 0 (Tor. Eagles) • Cell: City: Toronto → Link: /wiki/Toronto • Text: ...town of York..."
Reverse Grounding Fallback
• Maps answer back to graph coordinates • Direct synthesis if disconnected • Auditable evidence trail